Nearest Neighbors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Recall and choose your starting point

> Record your starting explanation for Nearest Neighbors before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Label Quality

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

Complete the supported Nearest Neighbors trace and record the first step you can explain without help.

Use W2: Apply **neighbour** and tie rules for Nearest **neighbour**s and **distance**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Nearest Neighbors explanation against the supplied facts before discussing it.

Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Training points: A length1 healthy, B length5 spotted, C length9 healthy. Query length7. Use absolute **distance**; for a nearest-**neighbor** tie choose the alphabetically earlier ID. Compare k=1 and k=3.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Nearest Neighbors target to a trace and a justified conclusion.

Training lengths 2(H),6(S),10(H); query=8. Compute **distance**s. Explain the tie and why combining length in mm with width in cm changes a two-feature **distance**.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Nearest Neighbors case and show the deciding evidence.

One-dimensional **training** points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute **distance**.

Find k=1 **neighbour**s and handle ties.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

For one feature, labeled points are 2:fern and 8:grass. A new point is 6. Under one-nearest-**neighbor** with absolute **distance**, predict its label.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 12 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Nearest Neighbors adds to Responsible Plant Classifier and keep the evidence.

Implement a transparent k-NN paper prototype for the Plant Classifier and record the chosen features, **scaling**, **distance**, and k.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Training lengths 2(H),6(S),10(H); query=8. Compute **distance**s. Explain the tie and why combining length in mm with width in cm changes a two-feature **distance**.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Calculate feature distances

> Calculate feature distances. Apply the supplied Nearest Neighbors facts and keep your reasoning.

For query 3 cm, calculate **distance**s to all five **training** rows. Explain why A and B tie even though their labels differ.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Apply neighbour and tie rules

> Apply neighbour and tie rules. Apply the supplied Nearest Neighbors facts and keep your reasoning.

For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Keep evaluation examples separate

> Keep evaluation examples separate. Apply the supplied Nearest Neighbors facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Nearest Neighbors", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Why must the three validation reference labels not enter the **neighbour** pool? Give the fixed k=3 score and baseline score on the worked validation examples.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Find neighbours independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Find neighbours independently. Apply the supplied Nearest Neighbors facts and keep your reasoning.

Rules supplied for this independent case

Rules: use absolute difference in centimetres. Rank neighbours by distance, then alphabetical ID, and use the majority among the selected k training labels. Fit the constant baseline to the training-majority label; a majority tie would choose clear. Do not add validation rows to the neighbour pool.

New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

My reasoning and evidence

W5 Interpret a vote and a match

> Interpret a vote and a match. Apply the supplied Nearest Neighbors facts and keep your reasoning.

Two of three **neighbour**s **vote** clear. Does that prove a two-thirds chance of a correct **prediction**? Does an exact feature match guarantee a correct label? Explain both limits.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Nearest Neighbors

> Let a nearest-neighbour picture show each distance before voting. Explain the labelled step before checking the changed condition.

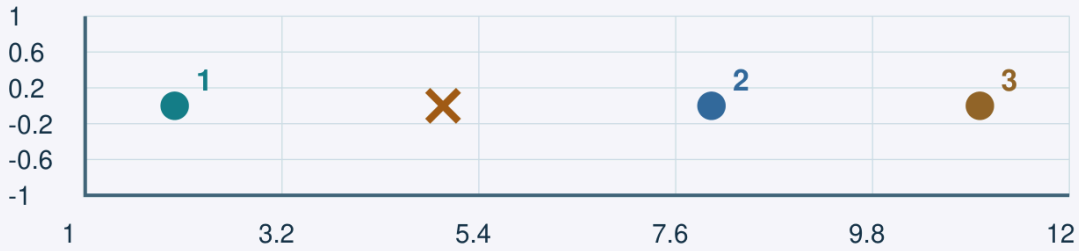
Let a nearest-**neighbour** picture show each **distance** before voting.

Training: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm. **Distance** is absolute difference; ties use alphabetically earlier ID.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/12

One-dimensional neighbours shown along length axis



Horizontal: length (cm); vertical: fixed display row, y=0

1: A healthy at (2, 0)

2: B spotted at (8, 0)

3: C spotted at (11, 0)

Cross: query at (5, 0)

Distance rule: Absolute length difference; ties use alphabetically earlier ID

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/12

Nearest Neighbors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Measure distances

To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm.

2. Apply k=1 rule

A and B tie nearest; declared ID tie-break chooses A, predicting healthy.

3. Compare k=3

All three vote: two spotted and one healthy, so k=3 predicts spotted.

CHANGE ONE THING

Move the query to length7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Let a nearest-**neighbour** picture show each **distance** before voting.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? All three **vote**: two spotted and one healthy, so k=3 predicts spotted.

1. Measure **distance**s

2. Compare k=3

3. Apply k=1 rule

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare k=3 All three **vote**: two spotted and one healthy, so k=3 predicts spotted. Read the picture in order.

All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move the query to length7 cm. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: k=1 predicts spotted from B, **distance**1; k=3 remains spotted. **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Repair a missing building block

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not **vote** away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Nearest Neighbors picture from its labels, then record a relevant limit.

01. Nearest Neighbors: the case

AI ACADEMY / AI-4/12 · 01 · LOOK AT THE FACTS

Nearest Neighbors: the case

The situation

Let a nearest-neighbour picture show each distance before voting.

Supplied evidence

Training: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm. Distance is absolute difference; ties use alphabetically earlier ID.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain nearest neighbors using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

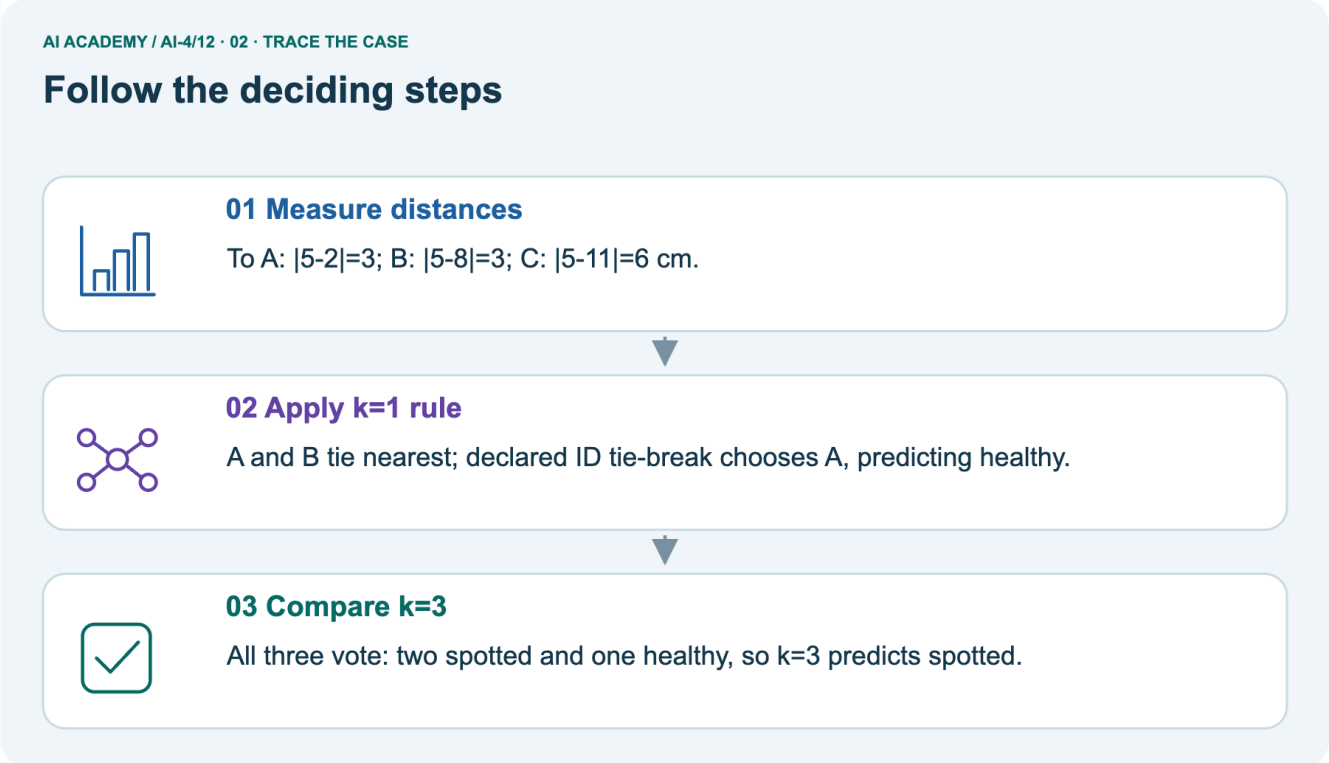
Let a nearest-**neighbour** picture show each **distance** before voting. Supplied facts: **Training**: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm.

Distance is absolute difference; ties use alphabetically earlier ID. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



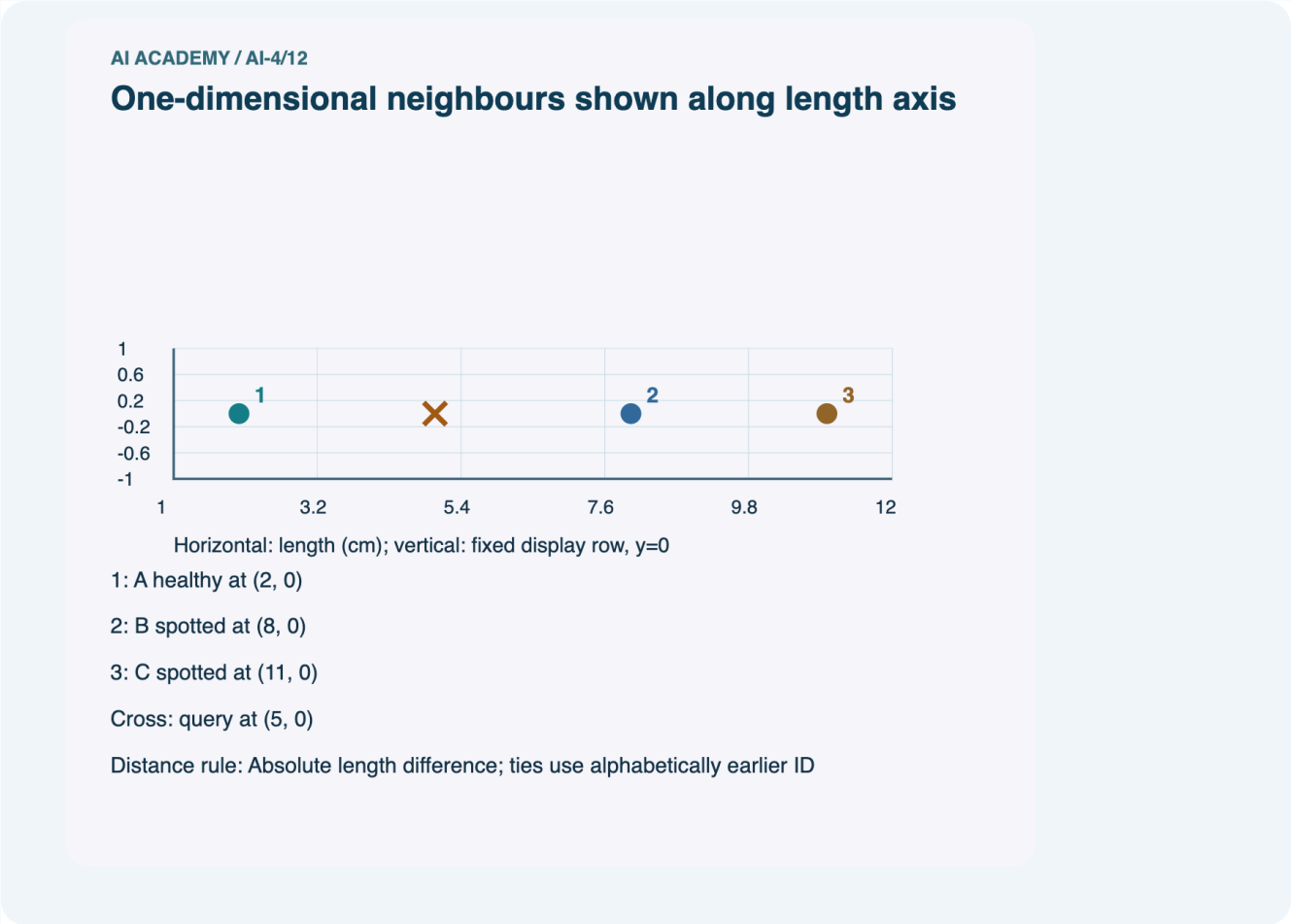
Follow the labelled picture with the matching explanation. Open the vector diagram

Measure **distance**s: To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm. → Apply k=1 rule: A and B tie nearest; declared ID tie-break chooses A, predicting healthy. → Compare k=3: All three **vote**: two spotted and one healthy, so k=3 predicts spotted.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. One-dimensional neighbours shown along length axis



Follow the labelled picture with the matching explanation. Open the vector diagram

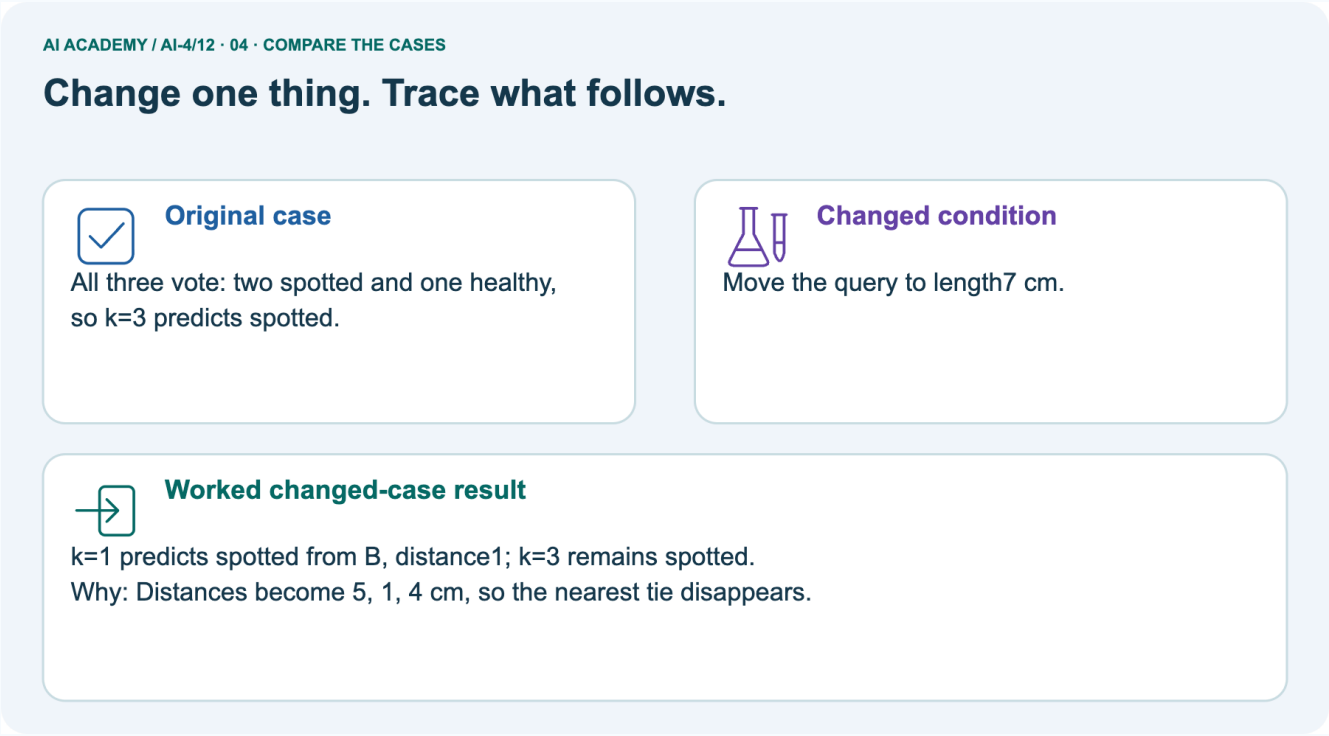
One-dimensional **neighbour**s shown along length axis. A healthy: (2, 0). B spotted: (8, 0). C spotted: (11, 0). Query query: (5, 0). Axes: length (cm); fixed display row, y=0.

Distance: Absolute length difference; ties use alphabetically earlier ID.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

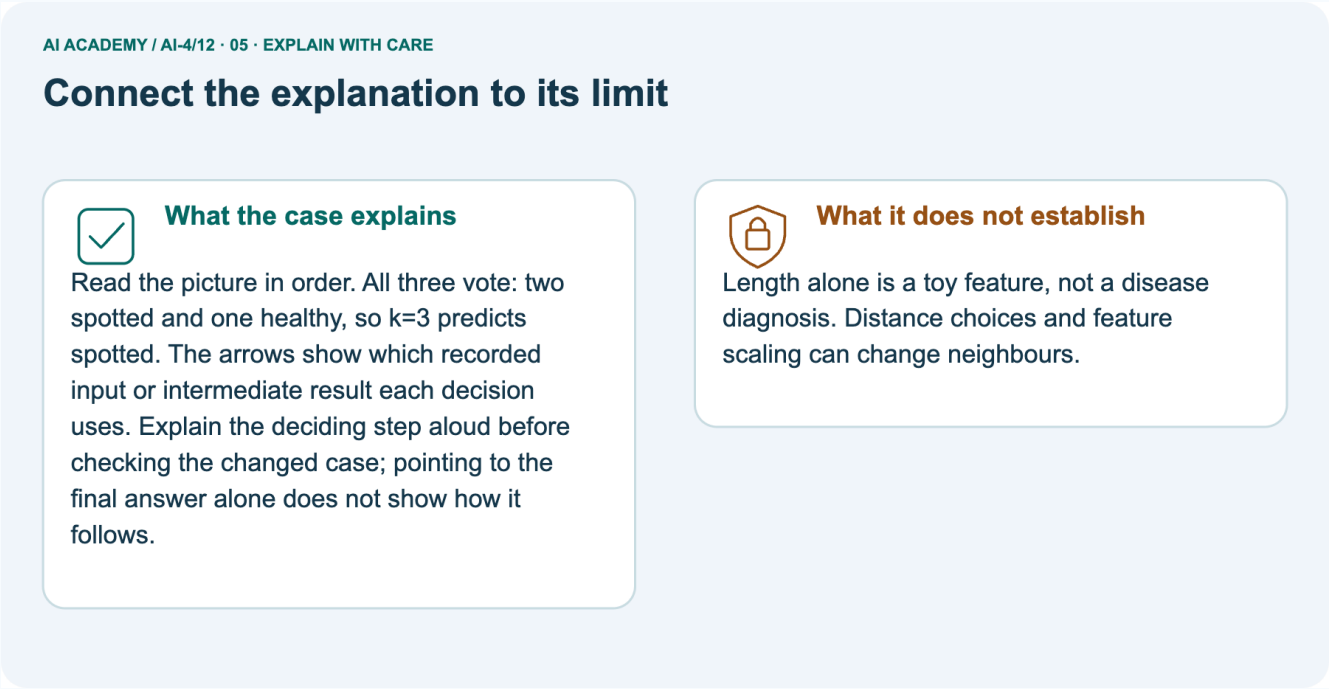
Original case: All three **vote**: two spotted and one healthy, so k=3 predicts spotted. Changed condition: Move the query to length7 cm. Expected result: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

Reason: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

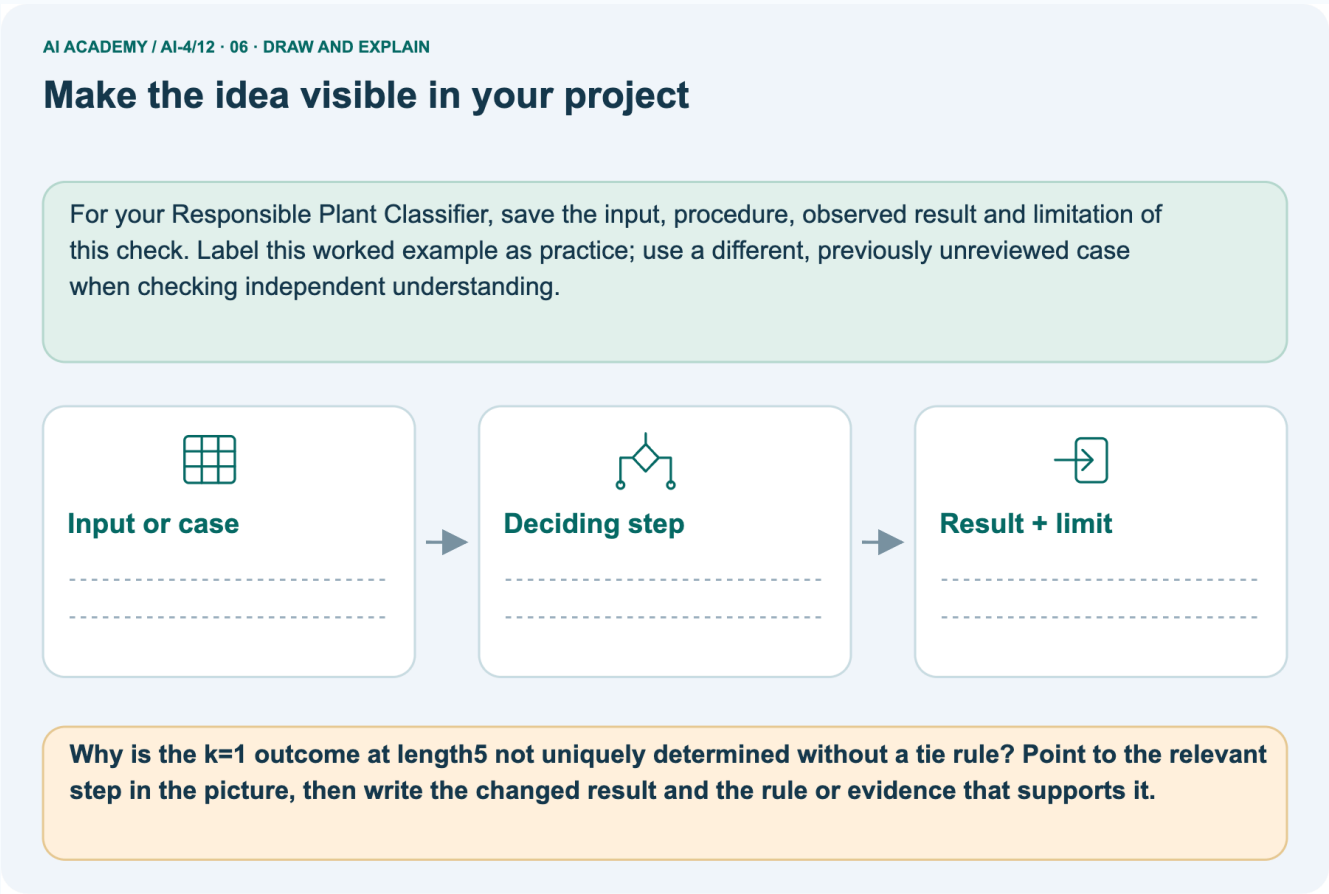
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Length alone is a toy feature, not a disease diagnosis.

Distance choices and feature **scaling** can change **neighbour**s.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.